01:XXX:XXX - Homework 3

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Assigned exercises: 1.1, 1.2, and 2.1–2.5 from *A Course in Regression and Smoothing Methods*.

Computational note: the numerical reference values below are those printed in the exercise, not independently executed results. The accompanying `homework3.R` implements all computations. Running it generates tables that this document automatically includes when they are available.

1. Exercise 1.1. For $x, a \in \mathbb{R}^p$ and a constant matrix $A \in \mathbb{R}^{p\times p}$, find the gradients of $a^T x$ and $x^T A x$.

Solution: (i) Linear form. Since $a^T x = \sum_{j=1}^{p} a_j x_j$, for each k,

$$\frac{\partial}{\partial x_k}(a^T x) = a_k.$$

Stacking the partial derivatives as a column vector gives

$$\boxed{\frac{\partial}{\partial x}(a^T x) = a.}$$

(ii) Quadratic form. Expand the scalar function as

$$x^T A x = \sum_{j=1}^{p}\sum_{k=1}^{p} A_{jk} x_j x_k.$$

For $\ell \in \{1, \ldots, p\}$, the product rule gives

$$\frac{\partial}{\partial x_\ell}(x^T A x) = \sum_{k=1}^{p} A_{\ell k} x_k + \sum_{j=1}^{p} A_{j\ell} x_j = (Ax)_\ell + (A^T x)_\ell.$$

Consequently,

$$\boxed{\frac{\partial}{\partial x}(x^T A x) = (A + A^T)x.}$$

If $A = A^T$, this reduces to $2Ax$.

For an alternative derivation, use differentials:

$$\begin{aligned} d(x^T A x) &= (dx)^T A x + x^T A\, dx \\ &= \big[(A + A^T)x\big]^T dx. \end{aligned}$$

Comparison with $df = (\nabla f)^T dx$ gives the same gradient. Only the symmetric part $A_s = (A + A^T)/2$ contributes to the quadratic form, since $x^T(A - A^T)x = 0$.

2. Exercise 1.2. Show that the two matrix-derivative expressions for the Hessian of a scalar function agree, and find the Hessian of $x^T A x$.

Solution: (i) Equality of the Hessian expressions. Assume h has continuous second partial derivatives in a neighborhood of x. Write

$$L = \frac{\partial}{\partial x^T}\left(\frac{\partial h}{\partial x}\right), \qquad R = \frac{\partial}{\partial x}\left(\frac{\partial h}{\partial x^T}\right).$$

By the derivative conventions in the exercise,

$$L_{jk} = \frac{\partial}{\partial x_k}\left(\frac{\partial h}{\partial x_j}\right), \qquad R_{jk} = \frac{\partial}{\partial x_j}\left(\frac{\partial h}{\partial x_k}\right).$$

Equality of mixed partials gives $L_{jk} = R_{jk}$ for every j, k. Therefore,

$$\boxed{\frac{\partial}{\partial x^T}\left(\frac{\partial h}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial h}{\partial x^T}\right) = \nabla^2 h(x).}$$

In particular, $\nabla^2 h(x)$ is symmetric. Merely assuming existence of second partials at a point would not, by itself, justify interchanging them; continuous second partials are a sufficient condition.

(ii) Quadratic-form Hessian. By Exercise 1.1,

$$\nabla(x^T A x) = (A + A^T)x.$$

Differentiating this linear vector-valued function yields

$$\boxed{\nabla^2(x^T A x) = A + A^T.}$$

If A is symmetric, the Hessian is $2A$.

3. Exercise 2.1. Find the minimum-variance unbiased estimator of the form $T_a = (1 - a)Z_1 + aZ_2$, where the independent variables satisfy $Z_j \sim N(\mu, \sigma_j^2)$ and $\sigma_j^2 > 0$.

Solution: For every fixed a,

$$E(T_a) = (1 - a)\mu + a\mu = \mu.$$

By independence,

$$V(a) = \mathrm{Var}(T_a) = (1 - a)^2\sigma_1^2 + a^2\sigma_2^2.$$

Differentiating,

$$V'(a) = -2(1 - a)\sigma_1^2 + 2a\sigma_2^2.$$

The first-order condition is

$$(1 - a)\sigma_1^2 = a\sigma_2^2,$$

so

$$\boxed{a^* = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}, \qquad 1 - a^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.}$$

Since $V''(a) = 2(\sigma_1^2 + \sigma_2^2) > 0$, this is the unique global minimum. Equivalently,

$$\boxed{T^* = \frac{\sigma_1^{-2} Z_1 + \sigma_2^{-2} Z_2}{\sigma_1^{-2} + \sigma_2^{-2}}.}$$

Substitution gives

$$\boxed{\mathrm{Var}(T^*) = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{1}{\sigma_1^{-2} + \sigma_2^{-2}}.}$$

Correction to the printed statement. For the estimator $(1 - a)Z_1 + aZ_2$, the correct proportionality is

$$(1 - a^*, a^*) \propto (\sigma_1^{-2}, \sigma_2^{-2}),$$

not $(a^*, 1 - a^*) \propto (\sigma_1^{-2}, \sigma_2^{-2})$. The latter would correspond to the estimator $aZ_1 + (1 - a)Z_2$.

If $\sigma_j^2 = c_j \sigma^2$, then

$$a^* = \frac{c_1}{c_1 + c_2}, \qquad 1 - a^* = \frac{c_2}{c_1 + c_2},$$

so the common unknown scale cancels. Normality is unnecessary for this variance-minimization argument; the common mean, independence, and finite positive variances suffice.

4. Exercise 2.2. Derive Newton–Raphson and Fisher scoring for probit regression.

Solution: Let $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$ independently, where

$$\eta_i = x_i^T \beta, \qquad p_i = \Phi(\eta_i), \qquad q_i = 1 - p_i, \qquad f_i = \phi(\eta_i).$$

Here Φ and ϕ are the standard normal CDF and density, and X has rows x_i^T.

Step 1: Log-likelihood and score. The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \{y_i \log p_i + (1 - y_i) \log q_i\}.$$

Since $dp_i/d\eta_i = f_i$,

$$u_i := \frac{\partial \ell_i}{\partial \eta_i} = \frac{y_i f_i}{p_i} - \frac{(1 - y_i) f_i}{q_i} = \frac{f_i (y_i - p_i)}{p_i q_i}.$$

Thus

$$\boxed{U(\beta) = \nabla \ell(\beta) = \sum_i x_i u_i = X^T u.}$$

Step 2: Observed information. Set $s_i = 2y_i - 1$, $t_i = s_i \eta_i$, and $r(t) = \phi(t)/\Phi(t)$. Then $\ell_i = \log \Phi(t_i)$ and

$$r'(t) = -r(t)\{t + r(t)\}, \qquad u_i = s_i r(t_i).$$

It follows that

$$w_i^{\text{obs}} = -\frac{\partial^2 \ell_i}{\partial \eta_i^2} = r(t_i)\{r(t_i) + t_i\}.$$

Equivalently,

$$w_i^{\text{obs}} = \begin{cases} f_i^2/p_i^2 + \eta_i f_i/p_i, & y_i = 1, \\ f_i^2/q_i^2 - \eta_i f_i/q_i, & y_i = 0. \end{cases}$$

Writing $W_{\text{obs}} = \text{diag}(w_i^{\text{obs}})$,

$$J(\beta) = -\nabla^2 \ell(\beta) = X^T W_{\text{obs}} X.$$

A first-order expansion of the score around $\beta^{(k)}$ gives

$$0 \approx U(\beta^{(k)}) - J(\beta^{(k)})(\beta^{(k+1)} - \beta^{(k)}).$$

Therefore the Newton–Raphson update is

$$\boxed{\beta^{(k+1)} = \beta^{(k)} + (X^T W_{\text{obs}}^{(k)} X)^{-1} X^T u^{(k)}.}$$

Step 3: Fisher information and scoring. Under the model, $E(u_i \mid x_i) = 0$ and

$$E(u_i^2 \mid x_i) = \frac{f_i^2}{p_i^2 q_i^2} \operatorname{Var}(Y_i \mid x_i) = \frac{f_i^2}{p_i q_i}.$$

Thus, with $w_i^F = f_i^2/(p_i q_i)$,

$$I(\beta) = X^T W_F X, \qquad W_F = \text{diag}(w_i^F).$$

Replacing observed information by expected information gives

$$\boxed{\beta^{(k+1)} = \beta^{(k)} + (X^T W_F^{(k)} X)^{-1} X^T u^{(k)}.}$$

Step 4: IRLS representation. Define the working response

$$z_i^{(k)} = \eta_i^{(k)} + \frac{y_i - p_i^{(k)}}{f_i^{(k)}}.$$

Since $w_i^F(z_i - \eta_i) = u_i$, scoring is equivalent to

$$\boxed{\beta^{(k+1)} = (X^T W_F^{(k)} X)^{-1} X^T W_F^{(k)} z^{(k)}.}$$

For probit regression, observed and expected information generally differ, so the two algorithms follow different paths but target the same finite MLE. In implementation, linear systems are solved instead of forming inverses.

5. Exercise 2.3. Implement logistic Newton–Raphson, logistic MM, probit Newton–Raphson, and probit Fisher scoring on `birthwt`; compare the prescribed starts and stopping rules.

Solution: Step 1: Data and design. Use the binary response `low` and the formula

$$\texttt{low} \sim \texttt{age+lwt+race+smoke+ptd+ht+ui+ftv}.$$

Race has reference level white; `ptd` $= 1\{$`ptl` $> 0\}$; and `ftv` is grouped into 0, 1, and 2+, with reference 0. This gives $n = 189$ observations and $p = 11$ coefficients, including the intercept. All four algorithms use the same design matrix.

Step 2: Logistic Newton–Raphson. For $p_i = \operatorname{expit}(x_i^T \beta)$,

$$\ell(\beta) = \sum_i \{y_i x_i^T \beta - \log(1 + e^{x_i^T \beta})\}, \quad U = X^T(y - p), \quad J = X^T W X,$$

where $W = \operatorname{diag}\{p_i(1 - p_i)\}$. Hence

$$\boxed{\beta^{(k+1)} = \beta^{(k)} + (X^T W^{(k)} X)^{-1} X^T (y - p^{(k)}).}$$

Step 3: Logistic quadratic MM. Let $L = -\ell$. Because $p_i(1 - p_i) \le 1/4$,

$$\nabla^2 L(\beta) \preceq \tfrac{1}{4} X^T X.$$

For any current iterate b, the second-order integral expansion gives

$$L(\beta) \le Q(\beta \mid b) := L(b) + \nabla L(b)^T (\beta - b) + \tfrac{1}{8} (\beta - b)^T X^T X (\beta - b).$$

The first-order condition for minimizing Q is

$$-X^T \{y - p(b)\} + \tfrac{1}{4} X^T X (\beta - b) = 0.$$

Therefore,

$$\boxed{\beta^{(k+1)} = \beta^{(k)} + 4(X^T X)^{-1} X^T (y - p^{(k)}).}$$

The majorization property gives

$$L(\beta^{(k+1)}) \le Q(\beta^{(k+1)} \mid \beta^{(k)}) \le Q(\beta^{(k)} \mid \beta^{(k)}) = L(\beta^{(k)}),$$

so the log-likelihood is nondecreasing. The two probit algorithms use the updates derived in Exercise 2.2.

Step 4: Initialization and convergence. The two starts are

$$\beta^{(0)} = 0, \qquad \beta^{(0)} = (X^T X)^{-1} X^T y.$$

The second is the least-squares fit to the binary response, used only as an initial value. For each start and method, use either

$$\|\beta^{(k)} - \beta^{(k-1)}\|_2 \le \epsilon$$

or

$$|K_n(\beta^{(k)}) - K_n(\beta^{(k-1)})| \le \epsilon, \qquad K_n = \ell/n,$$

with $\epsilon \in \{10^{-3}, 10^{-6}\}$. This produces $4 \times 2 \times 2 \times 2 = 32$ runs. The first coefficient update counts as iteration 1. The implementation uses full steps, without step-halving.

Step 5: Computational results. The accompanying R program reports the stopping iteration, log-likelihood, and all coefficient estimates for every run. The following tables are included automatically after running the script in the document directory.

Iteration table pending execution of the accompanying R script.

The 32-run coefficient table is generated by the same script.

Step 6: Interpretation. The logistic algorithms optimize the same likelihood, as do the two probit algorithms, so sufficiently accurate fits should agree within each link. MM uses a fixed upper bound on curvature and is generally slower near the optimum than Newton's method. Newton and scoring differ for probit because their curvature matrices differ. A tighter tolerance cannot give an earlier first stopping iteration along the same deterministic sequence of iterates. A least-squares start is not guaranteed to be better in every configuration. Finally, near a nonsingular maximizer,

$$\ell(\widehat{\beta}) - \ell(\beta) \approx \tfrac{1}{2}(\beta - \widehat{\beta})^T J(\widehat{\beta})(\beta - \widehat{\beta}),$$

so small likelihood changes need not imply equally small coefficient changes. These are theoretical expectations to compare with the generated output, not claims about unexecuted iteration counts.

6. Exercise 2.4. Reproduce the coefficient estimates, model-based standard errors, null deviance, and residual deviance for both links.

Solution: Step 1: Accurate coefficient fits. Use $\beta^{(0)} = 0$ and stop when

$$\|\beta^{(k)} - \beta^{(k-1)}\|_2 \le 10^{-10}.$$

Use Newton–Raphson for logit and Fisher scoring for probit.

Step 2: Model-based standard errors. The model-based covariance estimate is

$$\widehat{V}_{\text{model}} = I(\widehat{\beta})^{-1}, \qquad \widehat{\text{SE}}(\widehat{\beta}_j) = \sqrt{[\widehat{V}_{\text{model}}]_{jj}}.$$

For logit,

$$I(\widehat{\beta}) = X^T \operatorname{diag}\{\widehat{p}_i(1-\widehat{p}_i)\}X.$$

For probit,

$$I(\widehat{\beta}) = X^T \operatorname{diag}\left\{\frac{\phi(\widehat{\eta}_i)^2}{\widehat{p}_i(1-\widehat{p}_i)}\right\}X.$$

Expected information is used to reproduce the model-based probit standard errors from `glm`.

Step 3: Deviances. For individual Bernoulli observations, the saturated log-likelihood is zero, using $0 \log 0 = 0$. Thus

$$D_{\text{res}} = 2\{\ell_{\text{sat}} - \ell(\widehat{\beta})\} = -2\ell(\widehat{\beta}).$$

The intercept-only MLE has constant probability $\widehat{p}_0 = \overline{y}$, regardless of link. Hence

$$D_{\text{null}} = -2\sum_i \{y_i \log \overline{y} + (1-y_i)\log(1-\overline{y})\}.$$

The degrees of freedom are $n-1 = 188$ and $n-p = 178$, respectively.

Step 4: Reference values. The following are the values printed in the exercise, provided as benchmarks. Small last-digit differences may occur with a different convergence rule.

	Logit		Probit	
Term	Estimate	SE	Estimate	SE
Intercept	0.82302	1.24471	0.461585	0.722771
age	-0.03723	0.03870	-0.023300	0.022514
lwt	-0.01565	0.00708	-0.008925	0.004056
raceblack	1.19241	0.53597	0.704876	0.318383
raceother	0.74069	0.46174	0.441680	0.266840
smoke	0.75553	0.42502	0.464624	0.246865
ptdTRUE	1.34376	0.48062	0.816927	0.288368
ht	1.91317	0.72074	1.132272	0.426167
ui	0.68019	0.46434	0.414494	0.281152
ftv1	-0.43638	0.47939	-0.279556	0.279582
ftv2+	0.17901	0.45638	0.086412	0.269392

Quantity	Logit	Probit
Null deviance	234.67	234.67
Residual deviance	195.48	194.99
Null degrees of freedom	188	188
Residual degrees of freedom	178	178

The custom numerical estimates and standard errors are included with the robust-SE tables in Exercise 2.5 after running the R code.

7. Exercise 2.5. Compute model-robust standard errors for the logistic and probit coefficient estimates.

Solution: Step 1: Individual scores and asymptotic covariance. Let $s_i(\beta) = \partial \ell_i(\beta)/\partial \beta = x_i u_i(\beta)$. The likelihood estimator solves $\sum_i s_i(\widehat{\beta}) = 0$. Under possible misspecification, define $\beta^\dagger$ by $E\{s_i(\beta^\dagger)\} = 0$. With suitable regularity conditions,

$$\sqrt{n}(\widehat{\beta} - \beta^\dagger) \xrightarrow{d} N(0, A^{-1}BA^{-T}),$$

where

$$A = E\left\{-\frac{\partial s_i(\beta^\dagger)}{\partial \beta^T}\right\}, \qquad B = E\{s_i(\beta^\dagger)s_i(\beta^\dagger)^T\}.$$

This follows by Taylor expanding the estimating equation around $\beta^\dagger$, applying a law of large numbers to the score derivative, and a central limit theorem to $n^{-1/2}\sum_i s_i(\beta^\dagger)$.

Step 2: Empirical sandwich. Use the sum-based matrices

$$J = -\sum_i \frac{\partial s_i(\widehat{\beta})}{\partial \beta^T}, \qquad M = \sum_i s_i(\widehat{\beta})s_i(\widehat{\beta})^T.$$

The HC0 covariance estimate is

$$\boxed{\widehat{V}_{\mathrm{HC0}} = J^{-1}MJ^{-T}, \qquad \widehat{\mathrm{SE}}_{\mathrm{HC0}}(\widehat{\beta}_j) = \sqrt{[\widehat{V}_{\mathrm{HC0}}]_{jj}}.}$$

There is no additional factor of n because J and M are sums. Equivalently, with $\widehat{A} = J/n$ and $\widehat{B} = M/n$, $\widehat{V} = n^{-1}\widehat{A}^{-1}\widehat{B}\widehat{A}^{-T}$.

Step 3: Logistic case. Since $s_i = x_i(y_i - p_i)$,

$$J_L = X^T \operatorname{diag}\{\widehat{p}_i(1-\widehat{p}_i)\}X, \qquad M_L = X^T \operatorname{diag}\{(y_i - \widehat{p}_i)^2\}X.$$

Therefore,

$$\boxed{\widehat{V}_L = J_L^{-1}M_L J_L^{-1}.}$$

Step 4: Probit case. Here

$$\widehat{u}_i = \frac{\phi(\widehat{\eta}_i)(y_i - \widehat{p}_i)}{\widehat{p}_i(1-\widehat{p}_i)}, \qquad M_P = X^T \operatorname{diag}(\widehat{u}_i^2) X.$$

Using the observed weights from Exercise 2.2,

$$J_P = X^T \operatorname{diag}(\widehat{w}_i^{\text{obs}}) X, \qquad \boxed{\widehat{V}_P = J_P^{-1} M_P J_P^{-1}.}$$

For the model-robust probit sandwich, the bread is the empirical observed score derivative, not the model-expected information. Those population matrices need not agree under misspecification.

Step 5: Implementation and output. The R function `inference` constructs a matrix whose ith row is $s_i(\widehat{\beta})^T$, obtains M by its cross-product, and combines it with the observed information. It reports both HC0 and the optional correction

$$\widehat{V}_{\text{HC1}} = \frac{n}{n-p} \widehat{V}_{\text{HC0}}.$$

HC0 is the direct sandwich answer; HC1 is labeled separately.

Logit numerical inference table pending execution of the R script.

Probit numerical inference table pending execution of the R script.

Robust standard errors leave the coefficient estimates unchanged and need not exceed model-based standard errors. Under misspecification, they support inference for the pseudo-true coefficient, not an automatically causal parameter. The calculation assumes independent observations; clustering would require an appropriately clustered meat matrix.